SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

NDEJJE UNIVERSITY

KAMPALA CAMPUS

FACULTY OF SCIENCE AND COMPUTER STUDIES.

TINAH BRIDAL SPOT BOUTIQUE MANAGEMENT SYSTEM

COURSE UNIT: SOFTWARE PRINCIPLES 2

COURSE : BSE

YEAR : ONE

SEMESTER : TWO

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1. Overall description

1.1 Product perspective

The BMS is a standalone desktop application. It does not replace or integrate with any existing digital system because no such system currently exists at Tinah Bridal Spot. It is designed as a direct digital replacement for the physical transaction book and handwritten receipt book currently in use. All data will be stored locally on the boutique's computer. The system is self-contained and has no external dependencies beyond a standard Windows environment.

1.2 Product functions

At a high level, the BMS will perform the following functions:

- Record sales and rental transactions with customer details, item description, and amount paid.
- Automatically generate a numbered receipt for every transaction.
- Record business expenditures separately from sales income.
- Maintain an inventory list with real-time status of each item.
- Calculate and display a daily financial summary (income, expenditure, net balance).
- Allow the user to search transactions by customer name or date for dispute resolution.
- Calculate item selling price from import cost components.

1.3 User Characteristics

The primary users of the system are the business owner (Ms. Nankya Justine) and the shop employee (Ms. Nuwatekateka Yuditah). Both users are non-technical, with no prior experience using business management software. Both are comfortable using a basic computer for tasks such as watching CCTV footage. The system interface must therefore be simple, use plain language, and require minimal training. All labels and prompts must be in English without technical jargon

1.4 Constraints

- The system must be developed in C#.
- The system must run on a standard Windows PC without internet connectivity.
- The system must be operable by a non-technical user after a single demonstration session.
- All financial values are in Uganda Shillings (UGX).
- The development team consists of four first-year undergraduate students with foundational C# competency.

1.5 Assumptions and dependencies

- The boutique has access to a computer running Windows for deployment.
- A printer is available for physical receipt output if required.
- The system will be used for Tinah Bridal Spot only and not deployed elsewhere.
- Stakeholders are willing to adopt the system following a brief training session.
- Data entered into the system is assumed to be accurate at the time of entry.

2. Fuctional Requirements Specification

2.1 Table 1: Functional Requirements

Requirements ID	Requirements Description
FR-01	**Transaction Recording**
Precision	The system SHALL allow the user to record a new transaction by entering the customer name, item name, transaction type (sale or rental), quantity, unit price, and total amount paid. A transaction cannot be saved unless all mandatory fields are completed
Specificity	**Input:** Customer name (text), item name (text), transaction type (Sale / Rental), quantity (integer ≥ 1), unit price (UGX, positive number), amount paid (UGX, positive number). **Processing:** Validates that all mandatory fields are filled and that amount paid is greater than zero. Assigns a unique sequential transaction ID. Saves the record to the transaction log with the current date and time. **Output:** Transaction saved confirmation message displayed. Transaction ID assigned and visible to user. Record appears in the transaction log immediately. **Invalid Behavior:** If any mandatory field is empty: display error 'All fields are required.' If amount paid is zero or negative: display error 'Amount must be greater than zero.' Transaction is not saved until all errors are resolved.
FR-02	**Automated Receipt Generation**
Precision	The system SHALL automatically generate a numbered receipt for every successfully saved transaction. The receipt must display the shop name, receipt number, date, customer name, item, transaction type, amount paid, and any balance due.
Specificity	**Input:** Confirmed transaction record (from FR-01 or FR-08). **Processing:** Retrieves the saved transaction data. Assigns a sequential receipt number. Calculates balance due (unit price minus amount paid, if any). Formats and renders the receipt for display and optional printing. **Output:** Receipt displayed on screen within 3 seconds of transaction confirmation. Receipt number, date, item, amount, and balance due visible. Print option available. **Invalid Behavior:** If transaction data is incomplete or corrupted: display error 'Receipt could not be generated. Please check the transaction record.' Receipt number is not reused even if generation fails.
FR-03	**Expenditure Recording**
Precision	The system SHALL allow the user to record business expenditures separately from sales income. Each expenditure record must include a

	category, amount, and date. Expenditures must not be mixed with sales records in the transaction log.
Specificity	**Input:** Expenditure category (e.g. Tailor Fees, Food, Delivery, Other — selectable from a drop down), amount (UGX, positive number), date (defaults to current date, editable). **Processing:** Validates that category is selected and amount is greater than zero. Saves the expenditure to a separate expenditure log with a unique expenditure ID and timestamp. **Output:** Expenditure saved confirmation displayed. Record appears in the expenditure log immediately and is reflected in the daily financial summary. **Invalid Behavior:** If category is not selected: display error 'Please select an expenditure category.' If amount is zero or negative: display error 'Amount must be greater than zero.'
FR-04	**Daily Financial Summary**
Precision	The system SHALL calculate and display a daily financial summary for any selected date, showing total income from sales and rentals, total expenditure, and net balance (income minus expenditure). The summary must be generated on demand and must reflect all records for the selected date.
Specificity	**Input:** Date selection input (defaults to current date, user can select any past date). **Processing:** Queries all transaction records for the selected date and sums income. Queries all expenditure records for the selected date and sums expenditures. Calculates net balance = total income minus total expenditure. **Output:** Summary displayed showing: Total Income (UGX), Total Expenditure (UGX), Net Balance (UGX). Each figure is itemized to show contributing transactions. **Invalid Behavior:** If no records exist for the selected date: display message 'No transactions recorded for this date.' Net balance is shown as UGX 0 rather than an error.
FR-05	**Inventory Management**
Precision	The system SHALL maintain a list of stock items, each with a name, quantity, and current status (Available, Rented Out, or Sold). The system SHALL update an item's status automatically when a transaction involving that item is recorded.
Specificity	**Input:** For adding new stock: item name (text), quantity (integer ≥ 1), purchase cost components (see FR-07). For status updates: triggered automatically by FR-01 when a sale or rental transaction is saved. **Processing:** On new stock entry: saves item to inventory with status 'Available'. On sale transaction: reduces quantity by units sold; marks item as 'Sold' when quantity reaches zero. On rental transaction: marks item status as 'Rented Out'. On rental return (user action): resets status to 'Available'. **Output:** Inventory list updated immediately after any relevant transaction. Current status visible for each item. Item count reflects real-time

	stock levels. **Invalid Behavior:** If user attempts to record a transaction for an item with zero available quantity: display warning 'This item is not currently available in stock.' Transaction cannot be completed until stock is verified.
FR-06	**Transactions Search and Dispute Resolution**
Precision	The system SHALL allow the user to search the transaction log by customer name, date, or receipt number to retrieve a specific transaction record. This function exists specifically to support resolution of customer payment disputes.
Specificity	**Input:** Search parameter: customer name (partial text match accepted), or date (calendar picker), or receipt number (exact match). **Processing:** Queries the transaction log using the provided search parameter. Returns all matching records. Supports partial name matches (e.g. searching 'Namu' returns all records containing that string). **Output:** List of matching transaction records displayed, each showing receipt number, date, customer name, item, amount paid, and transaction type. User can select any record to view full details. **Invalid Behavior:** If no matching records found: display message 'No transactions found matching your search.' Search field remains active for a new query.
FR-07	**Item Price Calculator**
Precision	The system SHALL provide a price calculation tool that computes a recommended selling price for a new item based on its import cost components and a user-defined profit margin percentage.
Specificity	**Input:** Purchase cost (UGX), airfreight cost (UGX), importation tax (UGX), profit margin percentage (%, positive number entered by user). **Processing:** Calculates total cost = purchase cost + airfreight + tax. Calculates recommended price = total cost × (1 + profit margin / 100). Rounds result to the nearest 1,000 UGX. **Output:** Recommended selling price displayed (UGX). Breakdown of total cost and profit component shown. User can accept the price (which pre-fills the unit price field in FR-01) or enter a different price manually. **Invalid Behavior:** If any cost field is left empty or contains a non-numeric value: display error 'Please enter valid amounts for all cost fields.' If profit margin is zero or negative: display warning 'Profit margin should be greater than zero.'

3 Non-Functional Requirements Specifiction

2.2 *Table 2: Non-functional requirements*

NFR ID	Category	Requirement Statement	Acceptance Criterion
NFR-01	Usability	The system SHALL present all menus, labels, and prompts in plain English without technical jargon, such that a non-technical user can complete a full transaction cycle (record → receipt) without any assistance after a single 30-minute demonstration session.	A user with no prior software experience completes a transaction and generates a receipt within 3 minutes of first independent use, in 3 out of 3 observed test attempts.
NFR-02	Performance	The system SHALL generate and display a receipt within 3 seconds of the user confirming a transaction, measured on standard low-cost hardware (Intel Core i3 or equivalent, 4GB RAM, Windows 10).	Receipt is displayed within 3 seconds in 100% of tested transactions on the specified hardware configuration.
NFR-03	Reliability	The system SHALL not lose any recorded transaction or expenditure data if the application is closed unexpectedly, the computer is shut down without warning, or a power interruption occurs mid-session.	All transactions recorded in a session are fully retrievable after application restart in 100% of tested cases, including cases simulating abrupt shutdown.
NFR-04	Data Integrity	The system SHALL prevent the saving of any transaction or expenditure record with a zero, negative, or non-numeric amount by displaying a specific error message and blocking the save action.	Zero, negative, and non-numeric amount entries are rejected with a descriptive error message in 100% of tested cases. No invalid record is persisted to the data store.
NFR-05	Security	The system SHALL require the user to enter a 4-digit PIN before accessing any transaction records, financial summaries, or inventory data, to prevent unauthorized access by customers or visitors in the shop.	Unauthenticated access to any data screen is blocked in 100% of tested attempts. The system locks after 3 consecutive incorrect PIN entries and requires a restart to retry.
NFR-06	Maintainability	The system code SHALL be commented such that at least 80% of all methods contain a descriptive comment explaining the method's purpose, its input parameters, and its return value.	Static code review confirms that 80% or more of all methods contain compliant comments before Phase 4 submission.
NFR-07	Portability	The system SHALL be installable and fully operational on any Windows 10 or Windows 11 PC without requiring additional software	System installs and runs successfully on a clean Windows 10 and Windows 11 machine in test conditions, with no

		installations beyond the .NET runtime.	missing dependency errors.